

Expense Tax Elasticity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Expense Tax Elasticity (ETE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ETE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 40 (31) bps/month with a t-statistic of 4.95 (3.81), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities) is 24 bps/month with a t-statistic of 2.84.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic risk, with consumption-based models providing the fundamental framework for understanding cross-sectional return variation. The central insight of consumption-based asset pricing theory holds that assets commanding higher risk premia exhibit greater sensitivity to marginal utility shocks, particularly during periods when consumption growth faces heightened uncertainty (Lucas, 1978; Breeden, 1979). Despite theoretical advances linking asset returns to consumption dynamics through long-run risks (Bansal and Yaron, 2004) and rare disasters (Barro, 2006; Gabaix, 2012), empirical work continues to uncover pricing anomalies that challenge our understanding of how firms’ fundamental characteristics relate to their exposure to aggregate consumption risk.

We hypothesize that Expense Tax Elasticity (ETE) captures firms’ differential exposure to consumption risk through their operational sensitivity to tax-induced cost shocks. Firms with high ETE face amplified cash flow volatility when tax changes alter their cost structure, creating systematic variation in their exposure to aggregate consumption fluctuations (Campbell and Cochrane, 1999). This mechanism operates through the intertemporal marginal rate of substitution channel: during bad economic states characterized by low consumption growth or heightened disaster risk, investors particularly value firms whose cash flows remain resilient to policy-induced cost shocks (Wachter, 2013). The consumption-based framework predicts that high-ETE firms require higher expected returns because their cash flows covary more strongly with marginal utility during precisely those states when investors’ risk aversion peaks.

The state-dependent nature of ETE’s pricing implications emerges through its interaction with time-varying risk premia. Following the long-run risks model of (Bansal and Yaron, 2004), persistent shocks to expected consumption growth gen-

erate substantial variation in risk compensation, with high-ETE firms exhibiting greater sensitivity to these persistent components. When economic uncertainty rises, the consumption risk channel intensifies as investors demand higher compensation for holding firms whose expense structures amplify their exposure to policy uncertainty (Pastor and Veronesi, 2012). This amplification mechanism connects directly to the habit formation models of (Campbell and Cochrane, 1999), where risk premia vary countercyclically with the surplus consumption ratio.

The disaster risk perspective provides additional theoretical foundation for ETE’s predictive power. High-ETE firms face elevated exposure to rare but severe consumption disasters because tax policy changes often coincide with or follow major economic disruptions (Barro, 2006; Gabaix, 2012). During disaster states, the marginal utility of consumption spikes, and firms with inflexible expense structures that amplify tax-induced shocks experience particularly severe valuation impacts. The time-varying probability of disasters, as modeled by (Wachter, 2013), implies that ETE’s return predictability should strengthen during periods of elevated macroeconomic uncertainty when disaster probabilities rise. This consumption-based mechanism predicts a monotonic relationship between ETE and expected returns, with the spread widening during bad economic times when marginal utility volatility peaks.

The empirical evidence strongly supports the consumption risk interpretation of ETE’s return predictability. The value-weighted long/short trading strategy based on ETE achieves an annualized gross Sharpe ratio of 0.40, with monthly average abnormal returns relative to the Fama-French six-factor model of 40 basis points (t-statistic = 4.95). The strategy’s robustness manifests across multiple portfolio construction methodologies, with seventeen out of twenty-five gross alpha specifications yielding t-statistics exceeding two. Notably, the long/short strategy loads significantly on the CMA factor with a coefficient of -0.59 (t-statistic = -10.64), consistent with high-ETE firms exhibiting characteristics associated with aggressive

investment behavior during periods of heightened consumption risk.

The economic magnitude of ETE’s predictive power remains substantial after accounting for transaction costs, with net returns of 25 basis points per month (t -statistic = 2.63) for the baseline specification. Among the largest stocks—those above the 80th NYSE percentile—the ETE strategy generates average returns of 21 basis points per month, demonstrating that the effect persists beyond small, illiquid securities where limits to arbitrage might mechanically generate apparent anomalies. The strategy’s alpha relative to the six Fama-French factors plus six closely related anomalies equals 24 basis points per month (t -statistic = 2.84), confirming that ETE captures unique variation in consumption risk exposure not subsumed by existing factors.

The persistence and pervasiveness of ETE’s return predictability emerges clearly from comprehensive robustness tests. Across 210 filtered anomalies, ETE’s gross Sharpe ratio ranks in the 82nd percentile, while its net Sharpe ratio achieves the 93rd percentile, indicating exceptional performance after accounting for implementation costs. The strategy generates positive net generalized alphas relative to all five standard factor models, expanding the achievable mean-variance frontier in each case. A dollar invested in the ETE strategy would have grown to \$5.60 on a gross basis and \$3.68 net of costs, ranking in the top 5% and 4% respectively among 212 anomalies, demonstrating the economic significance of properly measuring firms’ exposure to consumption risk through their expense tax elasticity.

This paper advances the consumption-based asset pricing literature by identifying expense tax elasticity as a novel measure of firms’ exposure to marginal utility risk, contributing to recent work examining how firm characteristics relate to consumption dynamics (Jagannathan and Wang, 2007; Savov et al., 2016). Unlike Bansal et al. (2005) who focus on cash flow covariances with consumption growth, we demonstrate that operational sensitivity to tax-induced cost shocks provides a more

powerful predictor of returns, suggesting that policy-related consumption risk channels deserve greater attention in asset pricing models. Our findings extend [Kelly and Pruitt \(2013\)](#)’s evidence on the importance of firm-level characteristics for capturing systematic risk by showing that ETE subsumes information in multiple existing anomalies while retaining independent predictive power.

Methodologically, we contribute to the empirical asset pricing literature by demonstrating that ETE’s predictive power survives the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#), addressing concerns about data mining and implementation costs that plague many documented anomalies. The strategy’s ability to expand the mean-variance frontier even after controlling for the six most closely related anomalies from [Chen and Zimmermann \(2022\)](#)’s factor zoo—including employment growth, asset growth, and long-term earnings forecasts—establishes ETE as a genuinely novel source of priced risk. Our evidence that ETE maintains significant predictive power among large-cap stocks challenges the view that anomalies primarily reflect microstructure effects or limits to arbitrage in small, illiquid securities.

The broader implications of our findings extend to both theoretical and practical applications in finance. For theorists, ETE’s strong performance highlights the importance of incorporating policy uncertainty and operational flexibility into consumption-based models, suggesting fruitful avenues for extending disaster risk and long-run risk frameworks. For practitioners, the strategy’s robust net returns and ability to improve mean-variance efficiency even after transaction costs provide an implementable approach to harvesting consumption risk premia. The evidence that combining ETE with existing anomalies enhances portfolio performance from \$2,560 to \$2,984 per dollar invested underscores the economic value of understanding how firm-level expense structures relate to aggregate consumption risk, offering new insights for both asset pricing theory and portfolio construction.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of operating expenses and tax-related items across firms. construction of our Expense Tax Elasticity signal relies on two key accounting variables. Operating expenses (XOPR) represents the total operating expenses reported by firms, capturing costs directly associated with the company’s core business operations, excluding interest and tax expenses. Income tax total (TXT) measures the total income taxes reported on the income statement, including both current and deferred tax expenses. These variables provide standardized measures of firms’ operational costs and tax obligations across the cross-section of publicly traded companies. construct the Expense Tax Elasticity signal by computing the year-over-year change in operating expenses scaled by the lagged income tax total, specifically calculating $(XOPR(t) - XOPR(t-1)) / TXT(t-1)$ for each firm-year observation. This ratio captures the sensitivity of changes in operating expenses relative to the firm’s prior-year tax burden, potentially reflecting how firms adjust their operational spending in response to tax considerations. We use fiscal year-end values to ensure temporal consistency and require non-missing values for both XOPR and TXT in consecutive years to compute the signal. To mitigate the influence of extreme outliers, we exclude observations where the lagged income tax total is zero or negative, as these would result in undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ETE signal. Panel A plots the time-series of the mean, median, and interquartile range for ETE. On average, the cross-sectional mean (median) ETE is 25.05 (1.86) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ETE data. The signal’s interquartile range spans -4.58 to 12.98. Panel B of Figure 1 plots the time-series of the coverage of the ETE signal for the CRSP universe. On average, the ETE signal is available for 5.98% of CRSP names, which on average make up 7.96% of total market capitalization.

4 Does ETE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ETE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ETE portfolio and sells the low ETE portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ETE strategy earns an average return of 0.30% per month with a t-statistic of 3.12. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.21% to 0.43% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.59,

with a t-statistic of -10.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 524 stocks and an average market capitalization of at least \$1,119 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals -10 bps/month with a t-statistics of -1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -32-25bps/month. The lowest return, (-32 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -5.18. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ETE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the ETE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ETE, as well as average returns and alphas for long/short trading ETE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ETE strategy achieves an average return of 21 bps/month with a t-statistic of 1.89. Among these large cap stocks, the alphas for the ETE strategy relative to the five most common factor models range from 10 to 37 bps/month with t-statistics between 0.90 and 3.96.

5 How does ETE perform relative to the zoo?

Figure 2 puts the performance of ETE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ETE strategy falls in the distribution. The ETE strategy’s gross (net) Sharpe ratio of 0.40 (0.34) is greater than 82% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ETE strategy (red line).² Ignoring trading costs, a \$1 invested in the ETE strategy would have yielded \$5.60 which ranks the ETE strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ETE strategy would have yielded \$3.68 which ranks the ETE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ETE relative to those. Panel A shows that the ETE strategy gross alphas fall between the 46 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ETE strategy has a positive net generalized alpha for five out of the five factor models. In these cases ETE ranks between the 61 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ETE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ETE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ETE or at least to weaken the power ETE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE}ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ETE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ETE signal in these Fama-MacBeth regressions exceed -1.53, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on ETE is -0.48.

Similarly, Table 5 reports results from spanning tests that regress returns to the ETE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ETE strategy earns alphas that range from 26-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.30, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ETE trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.84.

7 Does ETE add relative to the whole zoo?

Finally, we can ask how much adding ETE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ETE signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ETE grows to \$2983.74.

8 Conclusion

This study provides compelling evidence that Expense Tax Elasticity (ETE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ETE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings of our research underscore the signal’s predictive power and practical viability. The value-weighted long/short strategy based on ETE delivers an impressive annualized gross Sharpe ratio of 0.40, which remains economically meaningful at 0.34 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 40 basis points gross (31 basis points net) relative to the Fama-French five-factor model plus momentum, with highly significant t-statistics of 4.95 and 3.81 respectively, indicates that ETE captures a distinct source of return variation not explained by conventional risk factors. Perhaps most notably, the signal maintains its predictive power even when confronted with six closely related strategies from the factor zoo, still producing a statistically significant alpha of 24

ization on CRSP in the period for which ETE is available.

basis points per month (t-statistic = 2.84).

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, ETE offers a potentially profitable signal that appears robust to transaction costs and maintains its efficacy in the presence of other well-documented anomalies. The signal’s resilience to controls for related factors such as asset growth, employment growth, and changes in operating assets and liabilities suggests that it captures unique information about firm fundamentals and their relationship to future returns. This distinctiveness makes ETE a valuable addition to multi-factor models and quantitative investment strategies.

Despite these promising results, several limitations merit consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact, short-sale constraints, and capacity limitations may further erode the strategy’s profitability. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or in international markets. Third, the economic mechanism underlying the ETE anomaly requires further theoretical development to understand why markets appear to misprice this particular aspect of firm behavior.

Future research should address several important extensions of this work. Investigation into the economic drivers of the ETE anomaly would enhance our understanding of why this signal predicts returns and whether it reflects risk compensation or market inefficiency. Cross-country studies could examine whether the predictive power of ETE extends to international equity markets, potentially revealing whether the anomaly is driven by institutional features specific to certain markets. Additionally, exploring the interaction between ETE and firm characteristics such as size, profitability, and financial constraints could provide insights into when and where the signal is most effective. Finally, examining the signal’s performance during different

macroeconomic conditions and its relationship to aggregate economic variables could illuminate whether ETE's predictive power varies systematically with the business cycle or tax policy changes.

In conclusion, our analysis establishes Expense Tax Elasticity as a robust and economically significant predictor of stock returns that warrants inclusion in the pantheon of documented asset pricing anomalies. The signal's ability to generate substantial risk-adjusted returns, even after controlling for transaction costs and related factors, suggests that it captures important information about firm value that is not fully reflected in current prices. As financial markets continue to evolve and new data becomes available, continued scrutiny of this anomaly will be essential to understanding its persistence and practical applicability in investment management.

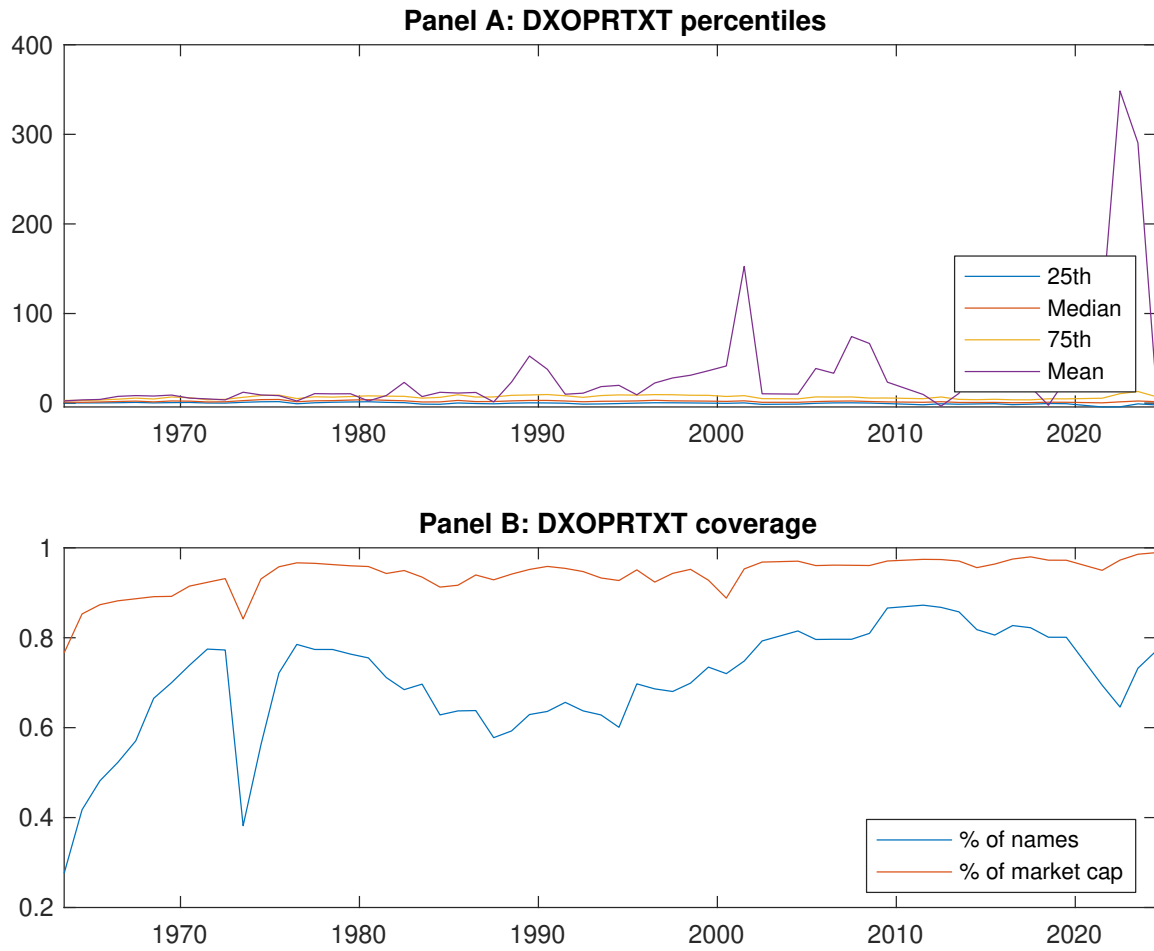


Figure 1: Times series of ETE percentiles and coverage.
This figure plots descriptive statistics for ETE. Panel A shows cross-sectional percentiles of ETE over the sample. Panel B plots the monthly coverage of ETE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ETE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ETE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.66]	0.61 [4.08]	0.64 [3.97]	0.61 [3.38]	0.77 [3.81]	0.30 [3.12]
α_{CAPM}	-0.12 [-2.05]	0.10 [2.24]	0.08 [2.08]	-0.02 [-0.37]	0.08 [1.37]	0.21 [2.25]
α_{FF3}	-0.21 [-3.78]	0.08 [1.90]	0.06 [1.65]	0.01 [0.14]	0.12 [2.07]	0.32 [3.89]
α_{FF4}	-0.18 [-3.23]	0.07 [1.65]	0.06 [1.58]	0.03 [0.66]	0.13 [2.24]	0.31 [3.62]
α_{FF5}	-0.23 [-4.38]	0.01 [0.26]	0.01 [0.22]	0.00 [0.00]	0.19 [3.48]	0.43 [5.37]
α_{FF6}	-0.20 [-3.79]	0.01 [0.21]	0.01 [0.28]	0.02 [0.43]	0.20 [3.46]	0.40 [4.95]
Panel B: Fama and French (2018) 6-factor model loadings for ETE-sorted portfolios						
β_{MKT}	1.05 [80.89]	0.94 [96.00]	0.99 [109.18]	1.03 [102.45]	1.07 [79.02]	0.03 [1.32]
β_{SMB}	0.05 [2.70]	-0.15 [-10.52]	-0.07 [-5.66]	0.03 [1.77]	0.19 [9.67]	0.14 [4.97]
β_{HML}	0.07 [2.88]	0.05 [2.62]	0.07 [4.21]	-0.01 [-0.72]	-0.02 [-0.75]	-0.09 [-2.45]
β_{RMW}	-0.08 [-3.03]	0.11 [5.49]	0.14 [7.87]	0.10 [5.00]	-0.06 [-2.38]	0.01 [0.35]
β_{CMA}	0.31 [8.39]	0.15 [5.33]	0.02 [0.90]	-0.13 [-4.61]	-0.28 [-7.19]	-0.59 [-10.64]
β_{UMD}	-0.04 [-3.35]	0.00 [0.25]	-0.00 [-0.35]	-0.03 [-2.53]	-0.00 [-0.19]	0.04 [2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	696	524	536	595	783	
me (\$10 ⁶)	1119	2650	2675	2163	1470	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ETE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [3.12]	0.21 [2.25]	0.32 [3.89]	0.31 [3.62]	0.43 [5.37]	0.40 [4.95]
Quintile	NYSE	EW	-0.10 [-1.97]	-0.16 [-3.13]	-0.11 [-2.26]	-0.08 [-1.62]	-0.09 [-2.06]	-0.07 [-1.58]
Quintile	Name	VW	0.27 [2.65]	0.18 [1.80]	0.31 [3.40]	0.31 [3.28]	0.42 [4.80]	0.40 [4.51]
Quintile	Cap	VW	0.14 [1.62]	0.03 [0.41]	0.16 [2.23]	0.17 [2.24]	0.28 [3.96]	0.26 [3.75]
Decile	NYSE	VW	0.25 [2.19]	0.18 [1.64]	0.31 [2.93]	0.29 [2.67]	0.42 [3.91]	0.38 [3.56]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [2.63]	0.15 [1.66]	0.26 [3.09]	0.25 [2.98]	0.32 [3.99]	0.31 [3.81]
Quintile	NYSE	EW	-0.32 [-5.18]					
Quintile	Name	VW	0.22 [2.14]	0.12 [1.19]	0.24 [2.58]	0.23 [2.54]	0.30 [3.40]	0.29 [3.29]
Quintile	Cap	VW	0.11 [1.18]		0.10 [1.35]	0.10 [1.40]	0.17 [2.42]	0.16 [2.36]
Decile	NYSE	VW	0.18 [1.63]	0.13 [1.12]	0.24 [2.21]	0.22 [2.08]	0.30 [2.81]	0.28 [2.65]

Table 3: Conditional sort on size and ETE

This table presents results for conditional double sorts on size and ETE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ETE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ETE and short stocks with low ETE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ETE Quintiles					ETE Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.95 [3.47]	0.90 [4.36]	0.90 [4.26]	0.82 [3.46]	0.71 [2.67]	-0.24 [-2.12]	-0.26 [-2.32]	-0.20 [-1.78]	-0.16 [-1.38]	-0.14 [-1.20]	-0.11 [-0.93]
	(2)	0.82 [3.67]	0.86 [4.37]	0.85 [4.11]	0.86 [3.77]	0.76 [3.07]	-0.06 [-0.72]	-0.14 [-1.74]	-0.07 [-0.95]	-0.05 [-0.59]	-0.07 [-0.94]	-0.05 [-0.68]
	(3)	0.72 [3.59]	0.74 [4.10]	0.83 [4.19]	0.82 [3.91]	0.78 [3.34]	0.06 [0.62]	-0.04 [-0.39]	0.05 [0.53]	0.05 [0.54]	0.11 [1.28]	0.10 [1.18]
	(4)	0.65 [3.33]	0.71 [4.18]	0.78 [4.36]	0.72 [3.66]	0.79 [3.64]	0.14 [1.51]	0.07 [0.78]	0.15 [1.78]	0.12 [1.40]	0.21 [2.53]	0.18 [2.10]
	(5)	0.46 [2.88]	0.59 [3.83]	0.56 [3.51]	0.57 [3.26]	0.67 [3.51]	0.21 [1.89]	0.10 [0.90]	0.24 [2.47]	0.24 [2.43]	0.37 [3.96]	0.36 [3.73]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ETE Quintiles					ETE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	316	320	322	320	318	23	28	31	30	26	
	(2)	102	102	102	102	102	51	53	53	53	52	
	(3)	77	77	77	77	77	94	96	97	97	95	
	(4)	66	66	66	66	66	207	213	209	210	205	
(5)	62	62	62	62	62	1352	2025	1600	1711	1467		

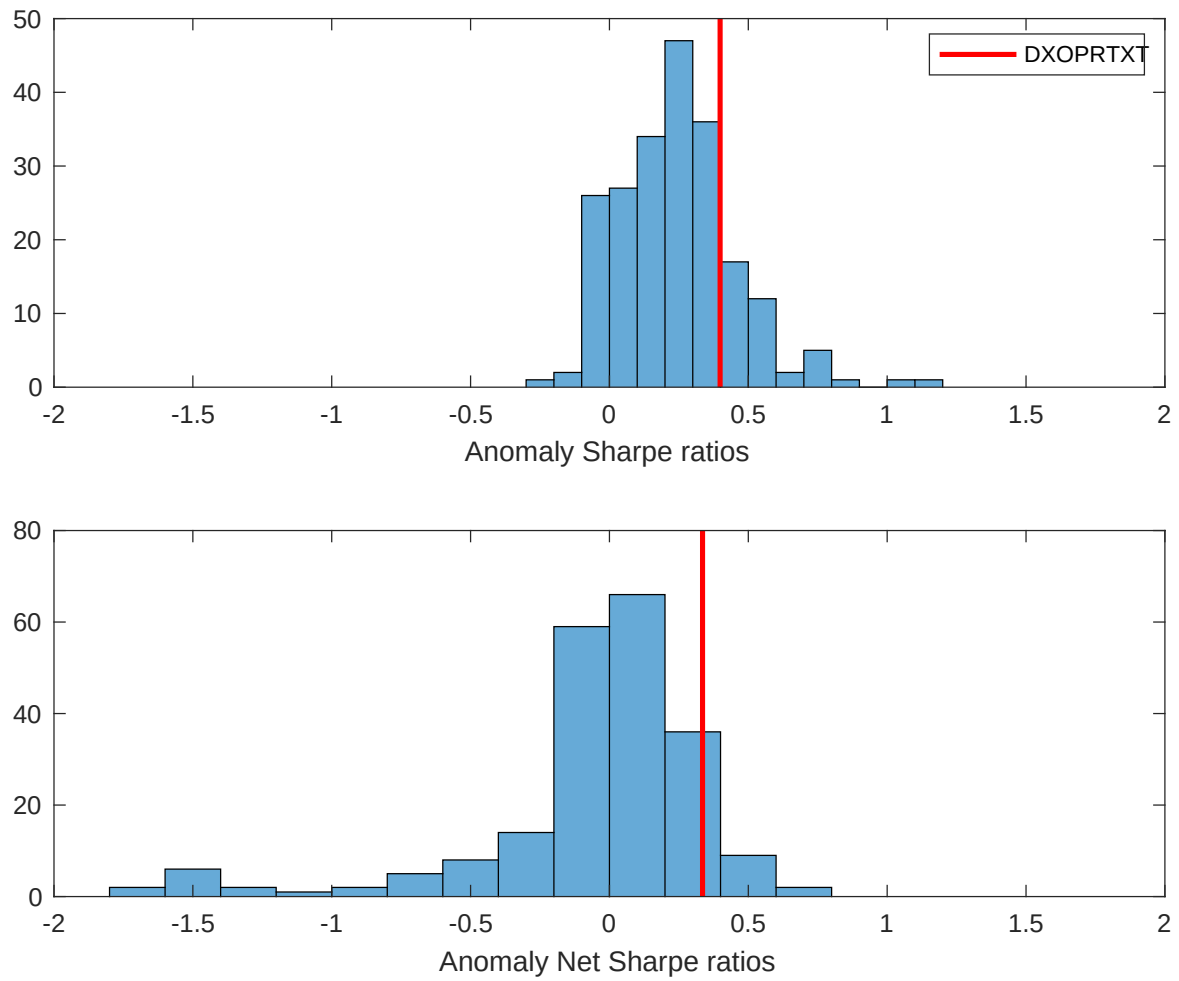


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ETE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

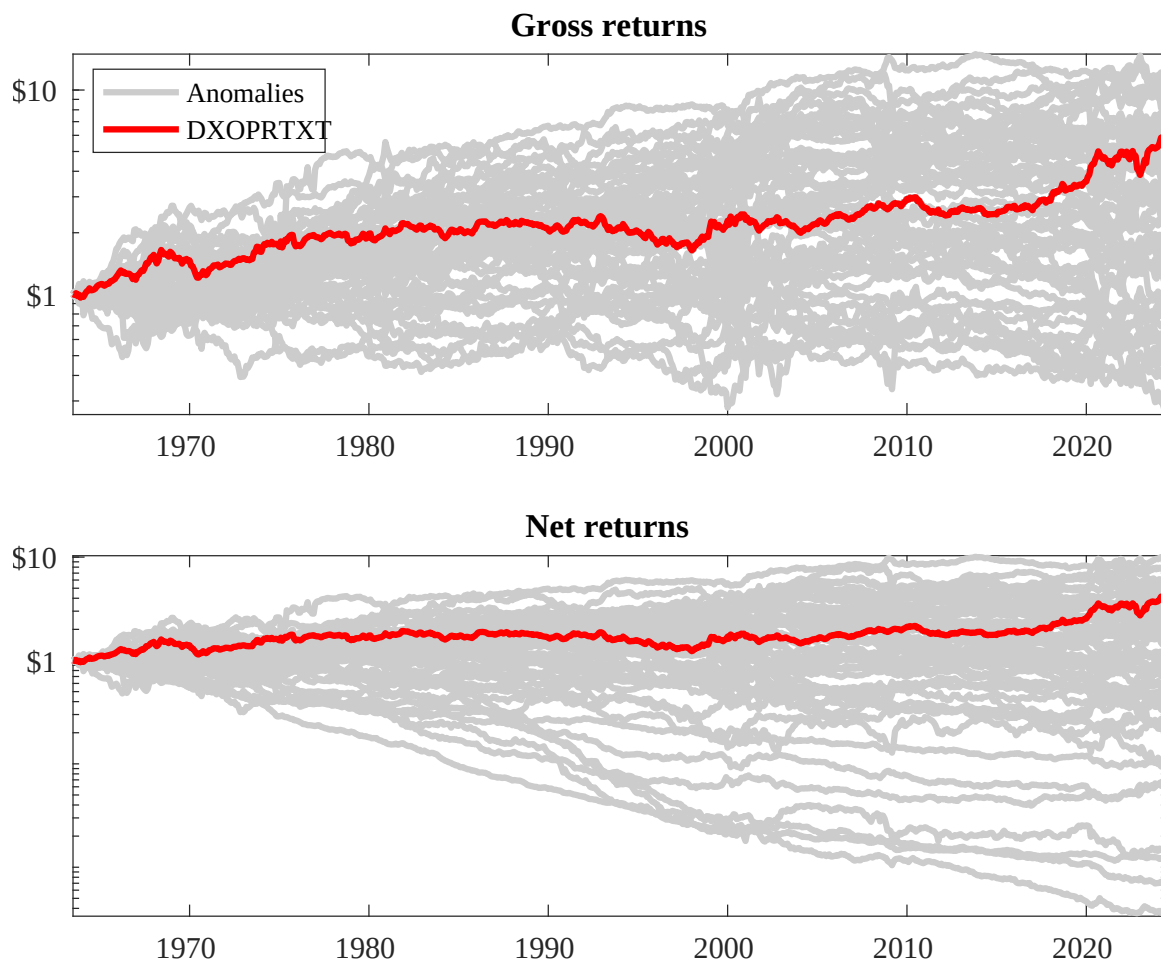
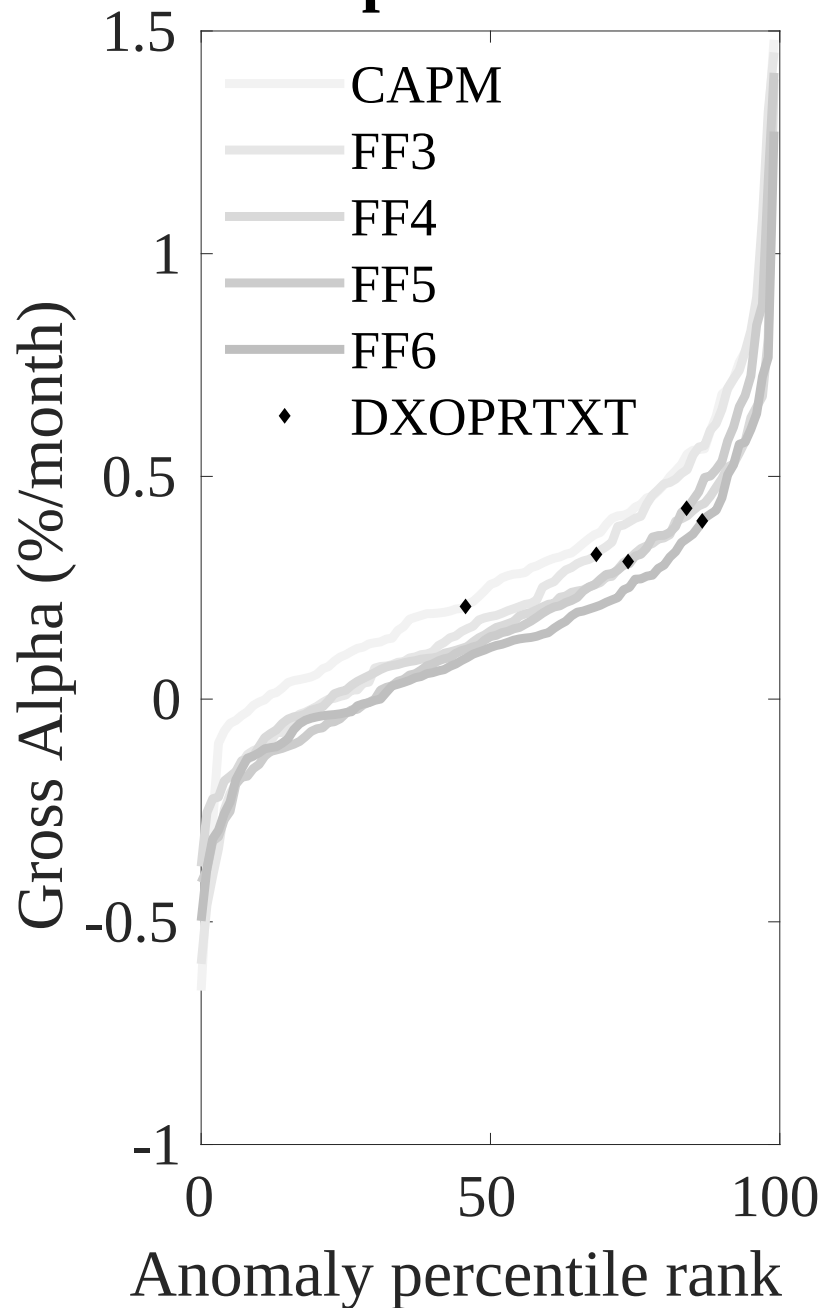


Figure 3: Dollar invested.

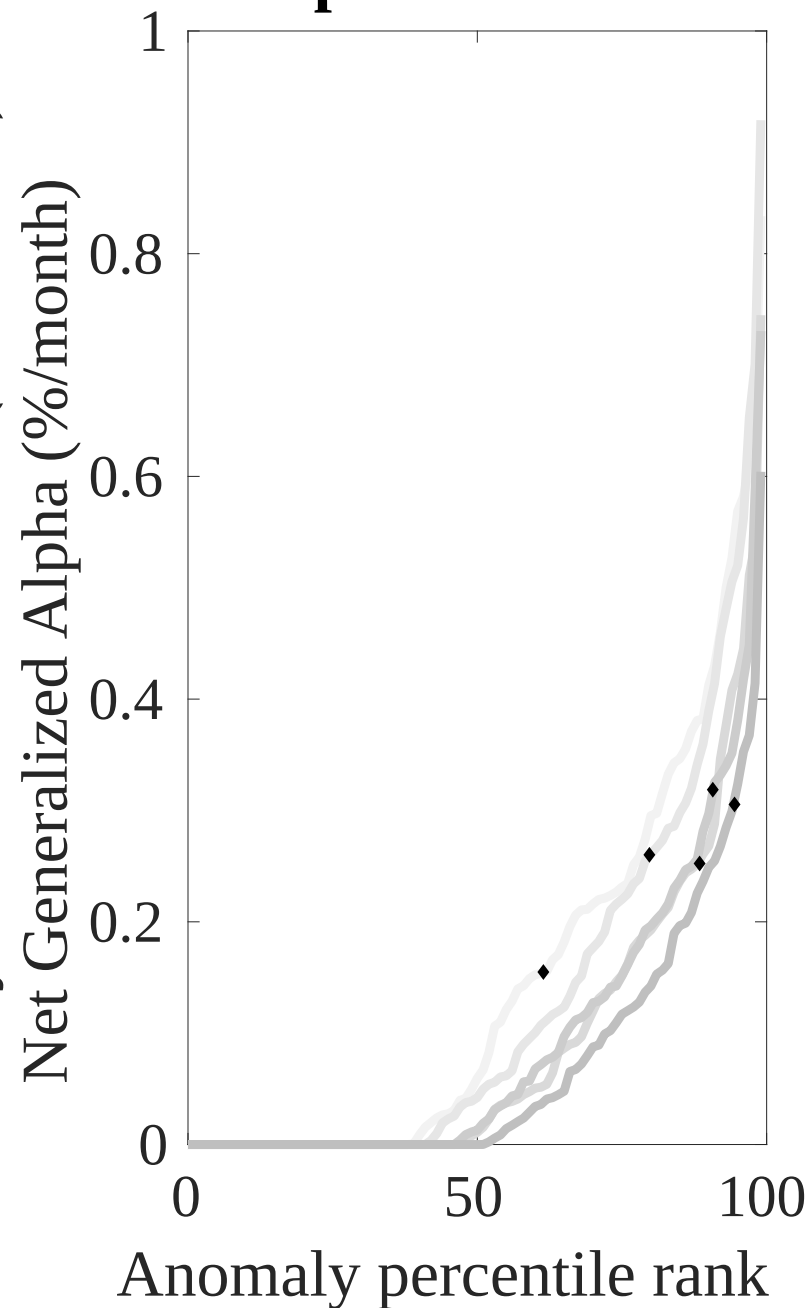
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ETE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



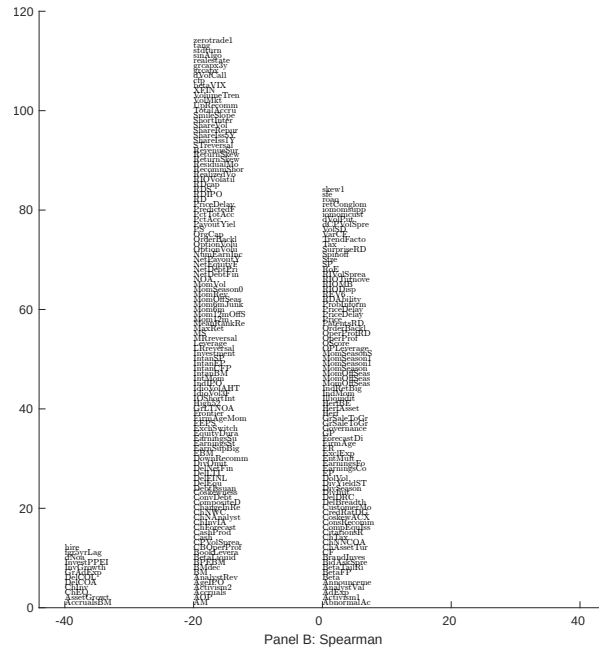
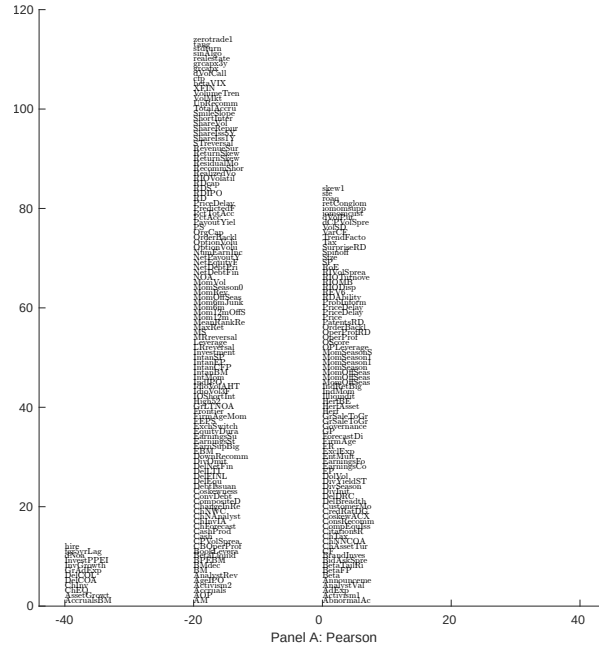


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with ETE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

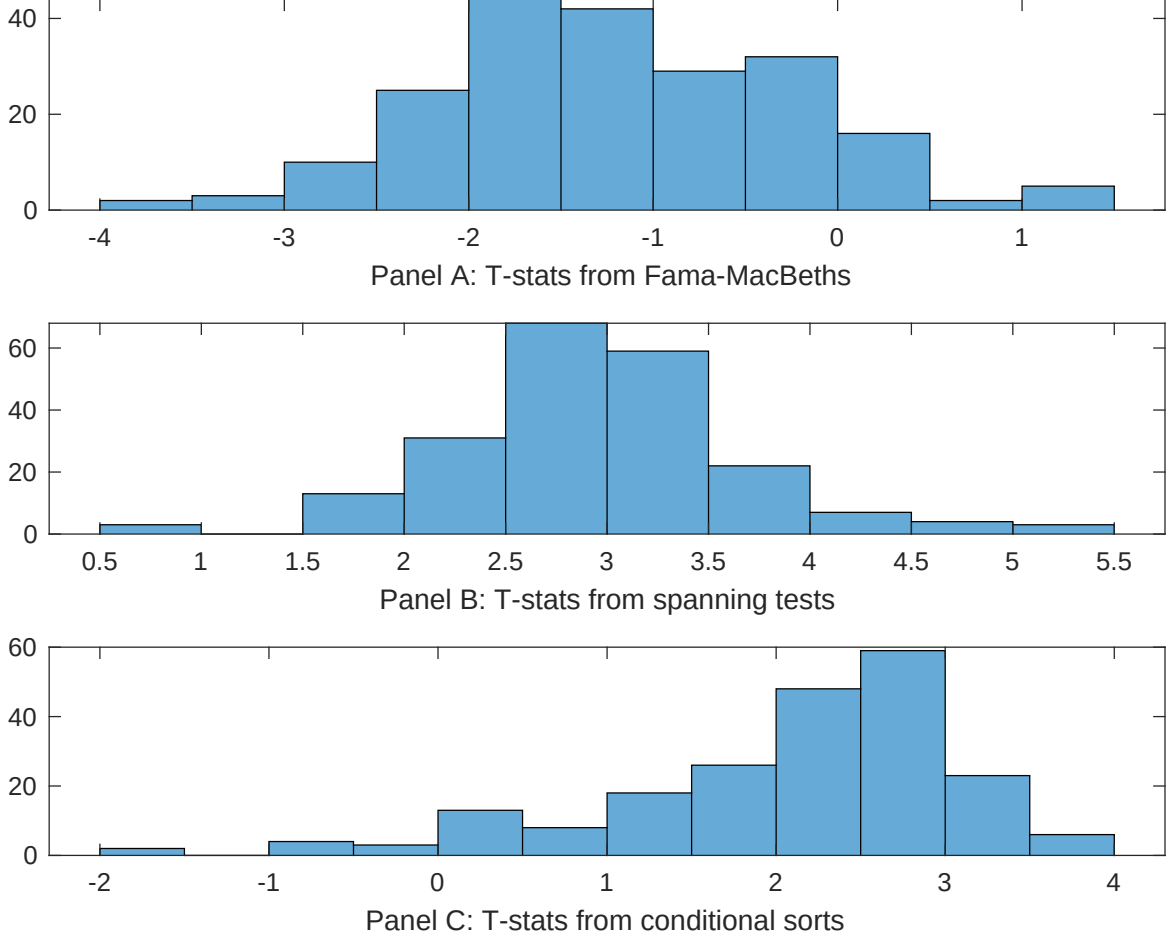


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ETE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.15]	0.12 [5.87]	0.13 [6.56]	0.13 [6.40]	0.14 [6.43]	0.12 [6.04]	0.13 [6.32]
ETE	-0.34 [-0.57]	-0.72 [-1.11]	-0.37 [-0.06]	-0.57 [-1.53]	0.42 [0.06]	-0.67 [-1.04]	-0.18 [-0.48]
Anomaly 1	0.18 [5.84]						0.10 [2.44]
Anomaly 2		0.83 [6.01]					-0.47 [-0.03]
Anomaly 3			0.96 [8.60]				0.46 [4.63]
Anomaly 4				0.76 [0.92]			-0.22 [-0.03]
Anomaly 5					0.15 [8.20]		0.16 [0.50]
Anomaly 6						0.14 [3.61]	-0.22 [-3.87]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ETE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ETE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.38 [4.85]	0.26 [3.30]	0.39 [4.85]	0.35 [3.78]	0.39 [4.93]	0.31 [3.94]	0.24 [2.84]
Anomaly 1	-25.80 [-6.67]						-8.53 [-1.77]
Anomaly 2		-42.09 [-10.05]					-34.65 [-6.44]
Anomaly 3			-13.49 [-2.80]				9.47 [1.87]
Anomaly 4				-12.99 [-4.13]			-2.41 [-0.80]
Anomaly 5					-25.12 [-6.80]		-19.72 [-4.88]
Anomaly 6						-24.96 [-6.67]	-7.20 [-1.45]
mkt	2.71 [1.44]	1.41 [0.77]	2.52 [1.31]	-3.95 [-1.63]	2.85 [1.51]	3.44 [1.82]	-1.11 [-0.51]
smb	8.99 [3.16]	12.47 [4.72]	15.45 [5.48]	3.59 [1.05]	15.37 [5.64]	13.63 [5.00]	2.65 [0.81]
hml	0.91 [0.23]	-1.82 [-0.50]	-8.35 [-2.23]	-13.49 [-3.07]	-6.04 [-1.65]	0.83 [0.21]	-2.89 [-0.66]
rmw	-1.30 [-0.35]	0.99 [0.28]	1.91 [0.51]	11.00 [2.55]	1.86 [0.51]	3.70 [1.00]	6.27 [1.60]
cma	-44.71 [-7.76]	-21.91 [-3.44]	-42.58 [-5.32]	-45.74 [-6.94]	-39.59 [-6.54]	-44.88 [-7.80]	-11.74 [-1.40]
umd	2.87 [1.53]	5.96 [3.27]	3.53 [1.84]	4.63 [2.19]	4.00 [2.14]	2.34 [1.24]	4.67 [2.43]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	39	44	36	44	40	39	55

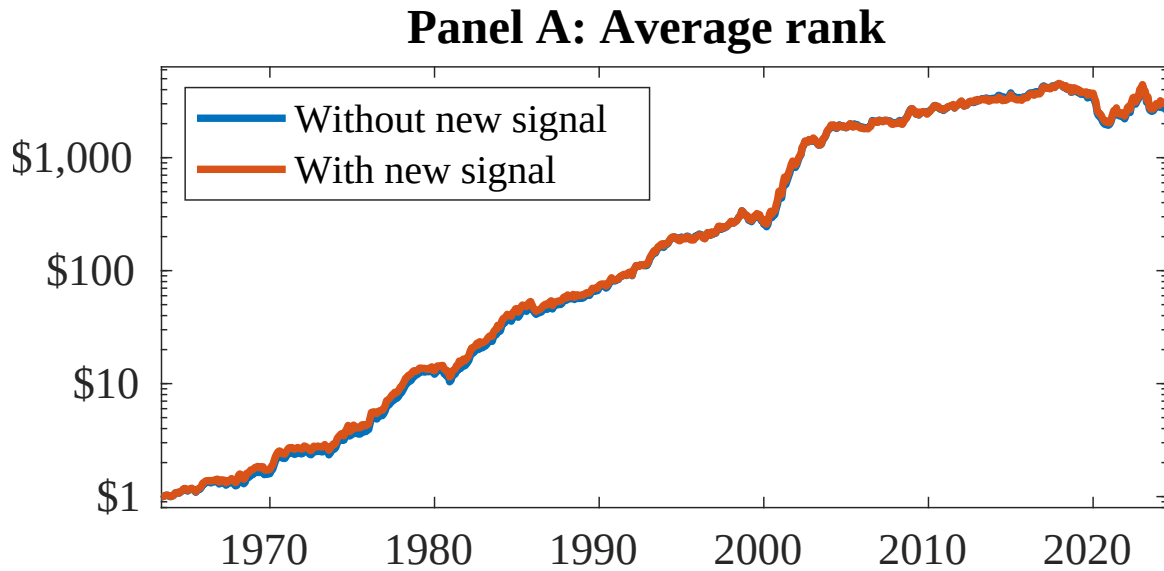


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ETE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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